09-04 第2篇

阅读理解（说明文） The Quiet Science Behind Every Rainbow

① Most of us have stopped walking at least once in our lives because of a rainbow. A long, curved bow of colour, drawn in the sky just after a shower, looks almost painted by hand. For thousands of years, people told stories about where rainbows came from. In China, a rainbow was sometimes said to be a bridge for spirits; in Europe, some children were told that a pot of gold lay at its foot. Today, science tells us something perhaps even more surprising: a rainbow is created by **very ordinary things** — sunlight and water drops — following very exact rules.

② To understand a rainbow, we must first understand **white sunlight**. At first glance, the light from the sun looks white, but it is in fact a mixture of many different colours of light, all travelling together. In **1666**, the English scientist **Isaac Newton** passed a narrow beam of white sunlight through a triangular piece of glass, called a **prism**, and found that the white beam split into the familiar band of **red, orange, yellow, green, blue, indigo and violet**. He proved that white light is not pure at all; it is a team of colours.

③ The second important idea is how light behaves when it meets water. Light travels fastest in air. When it enters water, it slows down slightly, and this change of speed makes the light **bend**. Scientists call this bending **refraction**. Different colours bend by slightly **different amounts**: red bends the least, violet bends the most. So when white sunlight enters a water drop, the mixed colours are **gently pulled apart**, like runners arriving at the finish line in slightly different order.

④ The third idea is called **reflection**. When light reaches the inside back wall of a water drop, it does not always pass straight through. A large part of it bounces back, as if the back of the drop were a small curved mirror. So inside a round raindrop, sunlight is first bent when it enters, then reflected from the back, and then bent again when it leaves.

⑤ Now we can put everything together. When the sun shines from behind you, and many small raindrops float in the air in front of you, each drop quietly does its work. Sunlight enters the drop, splits into colours by refraction, bounces off the back, and leaves the drop again. The red colour leaves a drop at one particular angle; the violet colour leaves at a slightly different angle. Your eyes, looking at millions of drops at the same time, see **red from the high drops** and **violet from the lower drops**. Between them, the other colours form the familiar curved bow.

⑥ Two everyday facts follow from this science. **First**, to see a rainbow, you must stand with your **back to the sun** and look towards the rain. No sun behind you, no rainbow in

front. **Second**, rainbows often appear in the **early morning** or **late afternoon**, when the sun is low in the sky. At noon, when the sun is nearly overhead, any rainbow would dip below the ground and be invisible to most people standing on flat land.

⑦ A rainbow is not actually a solid "object" at a fixed place in the sky. It is a **direction**. Your rainbow and your friend's rainbow, standing beside you, are made from different water drops. That is also why a rainbow seems to move when you walk towards it: each step, your eyes are catching light from a new set of drops. There is, sadly, no real "foot" where gold could be hidden.

⑧ Sometimes you can see a **second, fainter rainbow** just above the bright one. This second bow is formed when sunlight bounces **twice** inside each drop, rather than once. Because two reflections send the light out at a different angle, the colours of the second rainbow appear in **reverse order**: violet on top, red at the bottom. Pointing this out during a family walk is one of the small joys of learning a little science.

⑨ So next time the clouds break after a shower, look around carefully. Behind you, the sun should be low. In front of you, the air should still be full of tiny drops. If both are true, millions of round water drops are waiting to split, bounce and bend the light, ready to paint a curved bow just for your eyes. It is the same trick the sky has been quietly performing for billions of years — and you, standing there, are the only one who can see exactly **your** rainbow.

注释： prism 三棱镜；refraction 折射；reflection 反射；raindrop 雨滴；reverse 相反的

根据文章内容，选择最佳答案。

1. What did Isaac Newton prove with his prism experiment in 1666?

A. That a rainbow is only a story made up by children.
B. That water drops can disappear in sunlight.
C. That white sunlight is actually a mixture of many different colours.
D. That colours do not really exist in nature.

2. What is refraction, according to paragraph 3?

A. The complete stop of light inside a raindrop.
B. The bending of light when it enters a different material like water, with different colours bending different amounts.
C. The disappearance of colour when light meets water.
D. The way a rainbow takes a photograph.

3. Which colour leaves a drop at the highest angle, making it appear near the top of the main rainbow?

A. Violet.
B. Green.
C. Red.
D. Blue.

4. According to paragraph 6, where should you stand to see a rainbow?

A. With your face directly towards the sun.
B. With your back to the sun, looking towards the rain, usually early morning or late afternoon.
C. Exactly under the sun at noon.
D. Indoors with all curtains closed.

5. What is special about the second, fainter rainbow described in paragraph 8?

A. It has only one colour, red.
B. Its colours appear in reverse order, with violet on top and red at the bottom.
C. It always appears below the main rainbow.
D. It is made without any water drops at all.

答案与解析

1.
【答案】C
【题型】细节理解题
【定位】第②段 "He proved that white light is not pure at all; it is a team of colours"
【解析】牛顿的棱镜实验证明白光是多色混合。A、B、D 均无关。故选 C。

2.
【答案】B
【题型】词义理解/细节题
【定位】第③段 "this bending ... called refraction. Different colours bend by slightly different amounts"
【解析】折射即光进入不同介质时的弯折，且各色弯折角不同。A、C、D 均误读。故选 B。

3.
【答案】C
【题型】细节理解题
【定位】第⑤段 "red from the high drops and violet from the lower drops"
【解析】红色来自高处水滴，故出现在彩虹上方。A、B、D 均与原文不符。故选 C。

4.
【答案】B
【题型】细节理解题
【定位】第⑥段 "stand with your back to the sun and look towards the rain ... early

morning or late afternoon"
【解析】背对太阳、面向雨区，且多见于清晨或傍晚。其余均错。故选 B。

5.
【答案】B
【题型】细节理解题
【定位】第⑧段 "the colours of the second rainbow appear in reverse order: violet on top, red at the bottom"
【解析】副虹颜色反序排列。A、C、D 均与原文不符。故选 B。